

# Amir Beshir

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## Education

**Bachelor's Degree in Computer Science**

[American University of Ras Al Khaimah \(AURAK\)](#)

**08/2023 - 05/2027**

Ras Al Khaimah, UAE

**CGPA: 3.8**

**Modules:** Artificial Intelligence   Data Science   Statistics   Parallel Programming   Design & Analysis of Algorithms   OOP

## Research Experience

**Research Assistant**

[Advanced Technology and Artificial Intelligence Center](#)

**Supervisor:** [Prof. Khouloud Salameh](#)

**08/2026 - Present**

Ras Al Khaimah, UAE

**Fast Unsupervised SVG Image Retrieval using Clustering and Similarity-Based Representation**

- Investigated unsupervised and clustering techniques to improve similarity-based retrieval of panoramic dental X-ray images.
- Designed an efficient image representation and retrieval algorithm using geometric features and similarity measures.
- Evaluated the proposal against existing frameworks, analyzing retrieval quality, computational complexity and scalability.
- Conducted experimental analysis on dental X-ray SVG image data to identify effective feature combinations, clustering strategies and similarity measures for accurate and efficient image retrieval.

## Industry Experience

**Mobile Planning Intern**

[e& \(Etisalat\)](#)

**05/2026 - 07/2026**

Abu Dhabi, UAE

- Studied 5G network planning principles, focusing on coverage prediction, capacity planning and performance optimization.
- Evaluated antenna deployment strategies and radio propagation factors affecting network coverage and quality of service.
- Produced technical reports and presentations communicating engineering analyses and project findings to supervisors.
- Collaborated with planning engineers to understand data-driven approaches to mobile network design and optimization.

**Network & Security Technologies Intern**

[Alpha Data](#)

**05/2025 - 07/2025**

Abu Dhabi, UAE

- Monitored enterprise infrastructure using OpManager, Log360, Microsoft Defender, and LogRhythm SIEM to support network performance and security operations.
- Analyzed security alerts, system logs and incident data to identify potential threats and support cybersecurity investigations.
- Collaborated with security and network engineers to investigate operational issues, monitor infrastructure health and respond to security events.
- Prepared technical reports and presentations summarizing analyses, security observations, and outcomes for supervisors.

## Selected Projects — Full Portfolio: [amirbeshir.me](https://amirbeshir.me)

**[Student At-Risk Recovery Planner using A\\* Search](#)**

**03/2026 - 05/2026**

**Project:** Developed an AI-powered decision support system that generates personalized academic recovery plans using the A\* search algorithm with custom cost functions to identify optimal pathways toward graduation.

- Implemented the A\* algorithm from scratch, designing domain-specific heuristic functions and action costs to efficiently explore recovery strategies.
- Built an interactive Python dashboard for scenario analysis to evaluate recovery plans under different academic constraints.
- Compared A\* search against GBFS and UCS algorithms to evaluate solution quality and search efficiency.

**[Parallel Bank Transactions and Fraud Detection Simulation](#)**

**03/2026 - 05/2026**

**Project:** Developed a high-performance transaction processing simulator to evaluate serial, multithreaded, and multiprocessing architectures while detecting suspicious transaction patterns.

- Implemented serial, thread and process-based transaction processing systems using Python's threading and multiprocessing libraries.
- Benchmarked execution time, speedup, and scalability across multiple processing configurations.
- Designed fraud detection rules to identify suspicious transaction patterns during concurrent execution.
- Visualized performance metrics and analyzed synchronization challenges including race conditions and resource contention.

## Database Design and Implementation

10/2025 - 11/2025

**Project:** Designed and implemented a relational database system for a real-world application, applying database design principles, normalization, and SQL-based data management.

- Developed entity-relationship models and transformed them into normalized relational schemas up to 3NF.
- Implemented database tables, constraints, and SQL queries to support efficient data storage and retrieval.
- Incorporated data integrity, access control and privacy considerations following database design best practices.

## Campus Event Management System

05/2025 - 06/2025

**Project:** Developed a Java-based desktop application for managing university events, participant registration, and fee calculation using object-oriented software engineering principles.

**Code:** [Management System](#)

- Developed a modular Java Swing application via OOP principles including inheritance, polymorphism and encapsulation.
- Applied the Builder design pattern to create flexible event registration workflows with configurable participant options.
- Designed an intuitive graphical interface for role-based functionality, automated fee calculation and participant management.

## Network Analysis of Human Apoptosis Protein Interactions

04/2025 - 05/2025

**Project:** Applied graph theory and network analysis techniques to investigate the structural properties of human apoptosis protein-protein interaction networks using computational methods.

**Code:** [Protein Interactions](#)

- Constructed and analyzed biological interaction networks using Python and NetworkX.
- Identified influential proteins through graph-theoretic analysis and visualized interaction patterns via network visualizations.
- Interpreted characteristics of biological networks to demonstrate the application of computational methods in bioinformatics.

## Leadership and Outreach

### Student Government Association (SGA) Treasurer

American University of Ras Al Khaimah

08/2026 - Present

Ras Al Khaimah, UAE

- Managed SGA finances, including budgeting, fund allocation and account reconciliation, ensuring transparency.
- Maintained financial records, expenditure reports and audit documentation for student organizations and committees.
- Served on the Executive Board, collaborating on university wide events, initiatives and student programs.
- Presented financial updates and resource plans to the Executive Board and university administration.

## Key Skills

**Programming Skills:** • Python • C++ • Java • SQL • LaTeX

**Tools & Technologies:** • VS Code • MySQL • NI Multisim • Git • CMD

**Languages:** • English: Fluent • Arabic: Native

### Computer Science Skills:

• Cybersecurity • Network Security • Research • Operating Systems Analysis • Parallel Computing  
• Optimization • Data Structures • Digital Logic • Performance Benchmarking • Algorithm Analysis

### Soft Skills:

• Time Management • Communication • Adaptability • Problem-solving • Leadership  
• Attention to Detail • Critical Thinking • Work Ethic • Independence • Creativity

## Honors, Awards, and Recognition

- President's List (GPA 4.0/4.0) awarded for achieving a perfect academic record in Spring 2026.
- Dean's List (Fall 2023, Fall 2024, Spring 2025 and Fall 2026) for outstanding academic performance.
- Certificate of Completion in Advanced Algorithms and Complexity from UC San Diego through Coursera (2026).
- Certificate of Completion in AI Technology and Applications from Huawei ICT Academy (2026).
- Certificate of Completion in Operating Systems from Birla Institute of Technology & Science through Coursera (2026).
- Certificate of Completion in HCIA-Security V4.0 (Network Security) from Huawei ICT Academy (2025).
- 60% merit-based scholarship by AURAK (2026).
- Merit-based academic scholarship (50% waiver) awarded by AURAK (2025) and 40% merit-based scholarship (2023, 2024).